

PART A [25 marks]

This section consists of 3 questions. Answer **all** questions.

1 a) Find the following limits:

i) $\lim_{x \rightarrow \infty} \frac{2x^2 + x - 4}{1 - x^2}.$ [3 marks]

ii) $\lim_{x \rightarrow 2} \frac{3 - \sqrt{x+7}}{x^2 - 4}.$ [3 marks]

b) Consider a function $f(x) = \frac{1}{2 - \sqrt{x}}$. Find $\lim_{x \rightarrow \infty} f(x)$ and state the equation of horizontal asymptote for f . [3 marks]

2 Find $\frac{dy}{dx}$ for the following equations:

a) $y = 3^{2x+1}.$ [3 marks]

b) $e^{xy} + y = 5x.$ [4 marks]

c) $y = e^{2x} \tan x$ [4 marks]

3. A conical tank is of height 12 m and surface diameter 8 m. Water is pumped into the tank at the rate of $50 \text{ m}^3 / \text{min}$. How fast is the water level increasing when the depth of the water is 6 m?

[5 marks]

PART B [75 marks]

This section consists of 7 questions. Answer **all** questions.

1 a) Evaluate $\frac{1}{3-\sqrt{5}} - \frac{1}{1+\sqrt{5}}$. [3 marks]

b) Given $a = \log_x 2$ and $b = \log_x 3$, express $\log_x \left(\frac{9\sqrt{x}}{4} \right)$ in terms of a and b .

[3 marks]

c) Solve for p and q where $p \neq q$, such that

$$\frac{(p+qi)}{3i} = (3 + \sqrt{-16} - i^3).$$
 [4 marks]

2 a) Solve $|x^2 - x - 3| = 3$. [3 marks]

b) Solve the equation $3^{2x+1} - 28(3^x) + 9 = 0$. [4 marks]

c) Find the solution set for the inequality $2 - \left(\frac{x+2}{x-4} \right) < 5$. [4 mark]

3 a) A function f is defined by $f(x) = x^2 - 2x - 3$ for $0 \leq x \leq 5$. State the range of f and determine whether f is one to one. [5 marks]

b) Given $g(x) = \sqrt[3]{x}$ and $h(x) = \frac{1}{x^3}$. Find $f(x)$ such that $(f \circ g \circ h)(x) = \frac{x}{x+1}$.

[3 marks]

4. Given that $f(x) = \ln(2x-1)$.

a) Show that f is a one to one function. [2 marks]

b) Find the inverse of f . [2 marks]

- c) Find the domain and range for functions f and f^{-1} . [2 marks]
- d) Sketch the graphs f and f^{-1} on the same axes. [4 marks]
- 5 a) Determine the values of A and B such that the function
- $$h(x) = \begin{cases} Ax - B & , \quad x \leq -1 \\ 2x^2 + 3Ax + B & , \quad -1 < x \leq 1 \\ 4 & , \quad x \geq 1 \end{cases}$$
- is continuous for all values of x . [7 marks]
- b) Find $\lim_{x \rightarrow \infty} \frac{2x^2 + x - 4}{1 - x^2}$ [4 marks]
- 6 a) If $y = \sin(x^2 + 1)$, show that $x \frac{d^2 y}{dx^2} - \frac{dy}{dx} + 4x^3 y = 0$. [5 marks]
- b) Find the gradient of a curve $xe^{xy} = e^{2x} - e^{3y}$ at $(0, 0)$. [6 marks]
- c) If $y^2 - 2y\sqrt{1+x^2} + x^2 = 0$, show that $\frac{dy}{dx} = \frac{x}{\sqrt{1+x^2}}$. [5 marks]
7. A cylindrical container of radius r and h has a constant volume V . The cost of the materials for the surface of both of its ends is twice the cost of its sides. State h in terms of r and V . Hence, find h and r in terms of V such that the cost is minimum. [9 marks]

END OF QUESTION PAPER

SUGGESTED ANSWER**PART A**

1 a i) -2

ii) $-\frac{1}{24}$

b) 0 ; H.A : $y = 0$

2) a) $\frac{dy}{dx} = (2 \ln 3)3^{2x+1}$

b) $\frac{dy}{dx} = \frac{5 - ye^{xy}}{xe^{xy} + 1}$

c) $\frac{dy}{dx} = e^{2x}(\tan x + 1)^2$

3) $\frac{dh}{dt} = \frac{25}{2\pi} \text{ m/min}$

PART B

1 a) 1 b) $2b + \frac{1}{2} - 2a$ c) $p = -15$, $q = 9$

2 a) $x = -2$, $x = 0$, $x = 1$, $x = 3$

b) $x = -1$, $x = 2$

c) $\{ x : x < \frac{5}{2} \text{ or } x > 4 \}$

3 a) $R_f = [-4, 12]$; not one to one

b) $f(x) = \frac{1}{1+x}$

4. a) one to one

b) $f^{-1}(x) = \frac{e^x + 1}{2}$

c) $D_f = (\frac{1}{2}, \infty)$, $R_f = (-\infty, \infty)$; $D_{f^{-1}} = (-\infty, \infty)$, $R_{f^{-1}} = (\frac{1}{2}, \infty)$

5. a) $A = \frac{3}{4}$, $B = -\frac{1}{4}$

b) -2

6. b) $\frac{dy}{dx} = \frac{1}{3}$

7. $h = \frac{V}{\pi r^2}$; $r = \left(\frac{V}{4\pi} \right)^{\frac{1}{3}}$, $h = 1.721 V^{\frac{1}{3}}$